

LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Group Travel Expense Simplification Engine

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Section	I
Course	Software Engineering, PES University
Project	Group Travel Expense Simplification Engine (continuation of Lab 1)
Jira Space	Vib's (key: SCRUM), Scrum, company-managed

Structured Jira Evidence and Reflection Report

1. Lab Objective

The objective of this laboratory exercise was to apply Agile project-management practices using Jira. The functional and non-functional requirements captured in Lab 1 for the **Group Travel Expense Simplification Engine** were converted into a Jira backlog of Epics and User Stories. The exercise then covered prioritisation, Fibonacci-based story-point estimation, sprint planning, simulated sprint execution on a Scrum board, and interpretation of the Sprint Burndown Chart.

The project is a multi-currency expense-sharing utility. It records uneven group expenses, converts foreign-currency expenses into the trip's base currency using the day's exchange rate, and computes a minimum-transaction debt settlement plan so that a group of travellers can settle up with as few payments as possible. The actors identified in Lab 1 are the Group Traveler, the Trip Admin and the external Exchange Rate Service.

2. Backlog and Agile Planning

The five functional and two non-functional requirements from Lab 1 (FR-001 to FR-005, NFR-001 and NFR-002) were grouped into **four Epics** and broken down into **eight User Stories** written in the format *As a [role], I want [goal], so that [benefit]*. Each story title carries its Lab 1 requirement ID so that traceability back to the requirements table is preserved.

Priorities were assigned by user impact. The stories that form the core of the product (logging expenses, converting currency, viewing balances, computing the optimal settlement) and the two quality requirements were rated **High**. Trip-administration stories (FR-005, which Lab 1 rated Medium) were rated **Medium**. Story points follow the Fibonacci scale (2, 3, 5, 8) and reflect relative complexity, effort and uncertainty. The total backlog size is **31 story points**.

2.1 Epics

Epic	Theme	Lab 1 requirements	Stories	Points
Epic 1: Expense Logging & Currency Conversion	Recording group expenses and converting them to the base currency	FR-002, FR-003	SCRUM-12, SCRUM-13	8
Epic 2: Balance Tracking & Debt Settlement	Showing each traveller's balance and computing the minimum-transaction settlement	FR-004, FR-001	SCRUM-14, SCRUM-15	10

Epic	Theme	Lab 1 requirements	Stories	Points
Epic 3: Trip & Traveler Management	Admin control over travellers and the trip's base currency	FR-005	SCRUM-17, SCRUM-18	5
Epic 4: Performance & Reliability	Settlement speed and resilience when the rate feed is down	NFR-001, NFR-002	SCRUM-19, SCRUM-20	8
Total		7 requirements	8 stories	31

3. Jira Backlog Summary

ID	User Story	Lab 1 Req.	Epic	Priority	Pts
SCRUM-12	Log an Expense	FR-002	Epic 1	High	5
SCRUM-13	Convert Currency Automatically	FR-003	Epic 1	High	3
SCRUM-14	View My Balance	FR-004	Epic 2	High	2
SCRUM-15	See the Optimal Settlement Plan	FR-001	Epic 2	High	8
SCRUM-17	Set the Trip's Base Currency	FR-005	Epic 3	Medium	2
SCRUM-18	Manage Trip Travelers	FR-005	Epic 3	Medium	3
SCRUM-19	Fast Settlement Calculation	NFR-001	Epic 4	High	5
SCRUM-20	Keep Working When the Rate Feed Is Down	NFR-002	Epic 4	High	3
	Total				31

3.1 User story statements

SCRUM-12 Log an Expense (FR-002)

As a Group Traveler, I want to log an expense with its amount, currency, payer and the travellers who share it, so that the group's shared ledger always shows who paid for what.

SCRUM-13 Convert Currency Automatically (FR-003)

As a Group Traveler, I want my expense to be converted to the trip's base currency using that day's exchange rate, so that I can enter costs in whichever currency I actually paid in.

SCRUM-14 View My Balance (FR-004)

As a Group Traveler, I want to view my current balance at any time during the trip, so that I know how much I owe or am owed.

SCRUM-15 See the Optimal Settlement Plan (FR-001)

As a Group Traveler, I want the app to compute the smallest set of payments that settles the whole group, so that we avoid everyone paying everyone.

SCRUM-17 Set the Trip's Base Currency (FR-005)

As a Trip Admin, I want to set the trip's base currency, so that every expense is converted into one common currency.

SCRUM-18 Manage Trip Travelers (FR-005)

As a Trip Admin, I want to add or remove travellers from a trip, so that the group list stays accurate when plans change mid-trip.

SCRUM-19 Fast Settlement Calculation (NFR-001)

As a Group Traveler, I want the settlement to be calculated in under 100 ms for up to 50 travellers and 500 expenses, so that checking who owes what never feels slow on mobile data.

SCRUM-20 Keep Working When the Rate Feed Is Down (NFR-002)

As a Group Traveler, I want expense logging to continue using the last known exchange rate when the live feed is unavailable, so that I can keep recording expenses with poor connectivity.

4. Sprint Planning and Execution

Sprints were created in Jira with a one-week duration (9 Sep to 15 Sep 2026). Because the full backlog (31 points) is larger than what a single week should hold, stories were selected for the sprint by combining value and size: **SCRUM-15** (8 pts, High, the core min-flow settlement feature), **SCRUM-20** (3 pts, High, rate-feed resilience) and **SCRUM-18** (3 pts, Medium, traveller management). Together they make a sprint of **14 story points** across three epics. The remaining five stories (17 points) were left in the product backlog for later sprints.

During execution the stories were moved across the Scrum board columns **To Do, In Progress and Done**. The Sprint 1 board screenshots capture three states: all stories waiting in To Do, work in progress with SCRUM-18 and SCRUM-15 started while SCRUM-20 was still waiting, and finally all three stories in Done. For Sprint 2, the backlog view shows the SCRUM Sprint 2 container holding SCRUM-15, SCRUM-18 and SCRUM-20 with its own estimate, and the Sprint 2 board shows the work in flight (SCRUM-15 and SCRUM-20 In Progress, SCRUM-18 still To Do).

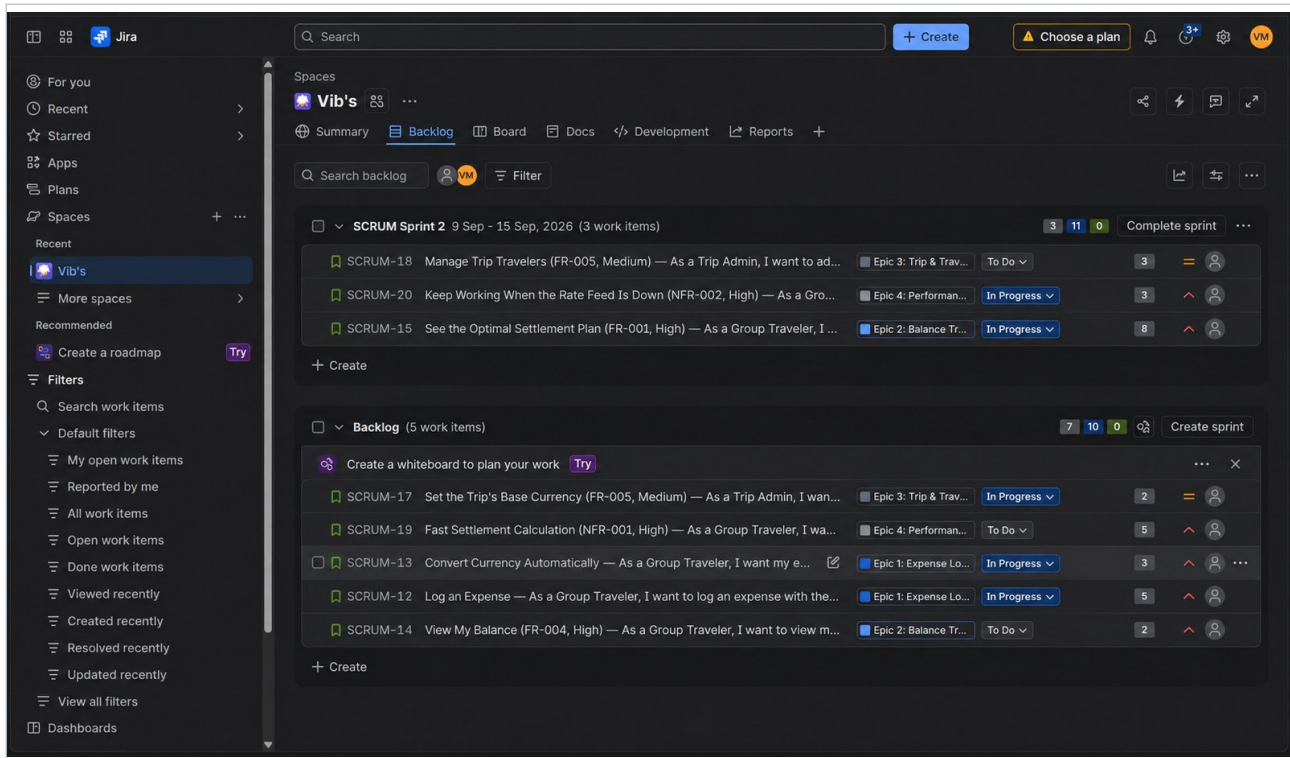
After each sprint the Burndown Chart was opened from *Reports* to compare the remaining story points with the guideline (ideal burn rate).

5. Jira Evidence Screenshots

The following screenshots are the visual evidence for this submission. Each caption explains what the screen demonstrates.

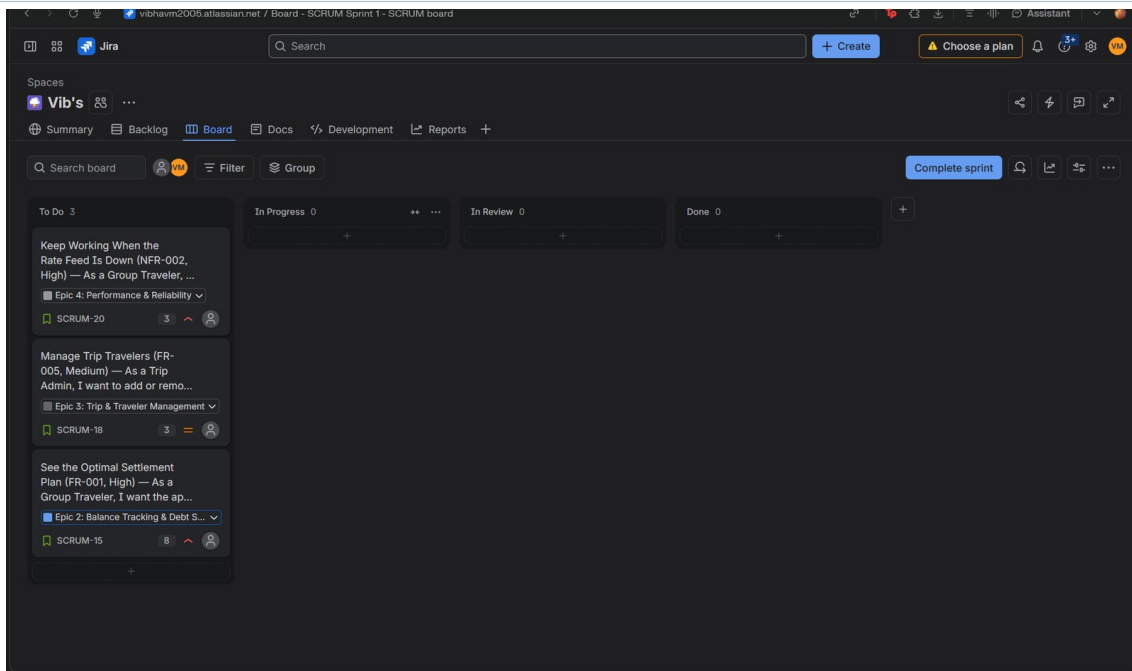
1. Jira backlog with Epics, User Stories, priorities and story points

The backlog view shows SCRUM Sprint 2 (9 Sep to 15 Sep 2026) containing SCRUM-18, SCRUM-20 and SCRUM-15 (14 points), and the product backlog holding the other five stories SCRUM-17, 19, 13, 12 and 14 (17 points). Every story displays its Epic label, status, story-point estimate (2, 3, 5 or 8) and priority icon, which also serves as the story-point assignment evidence.



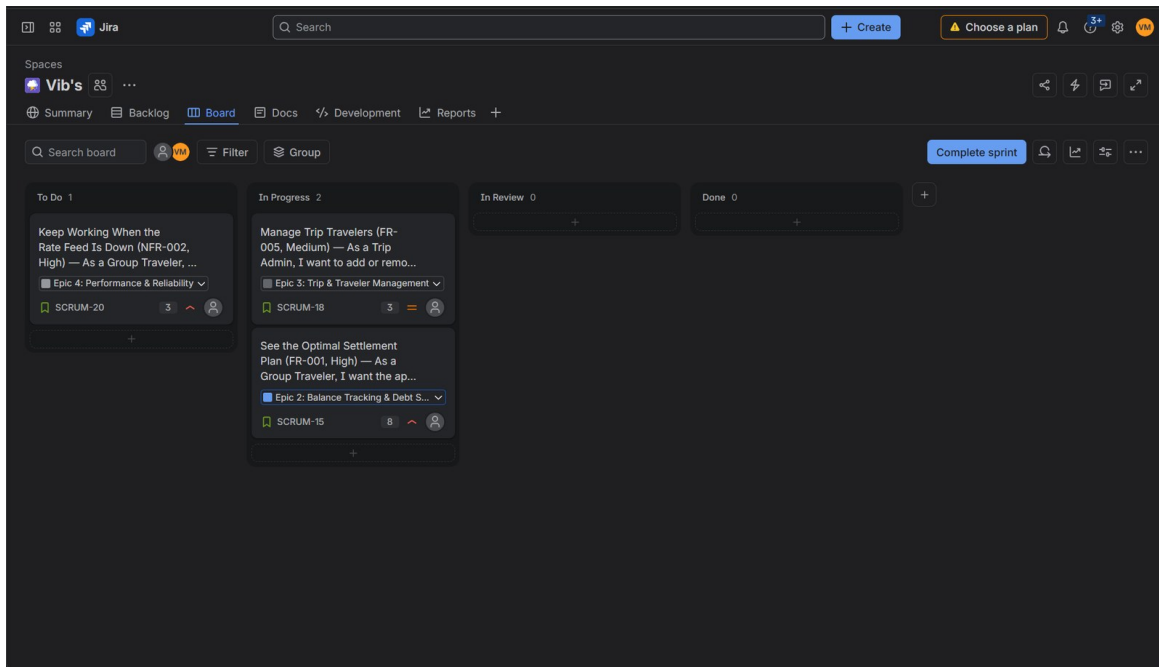
2. Sprint 1 board - sprint started (To Do)

The Scrum board right after the sprint was started: SCRUM-20 (3 pts), SCRUM-18 (3 pts) and SCRUM-15 (8 pts) are all in To Do, with In Progress, In Review and Done empty.



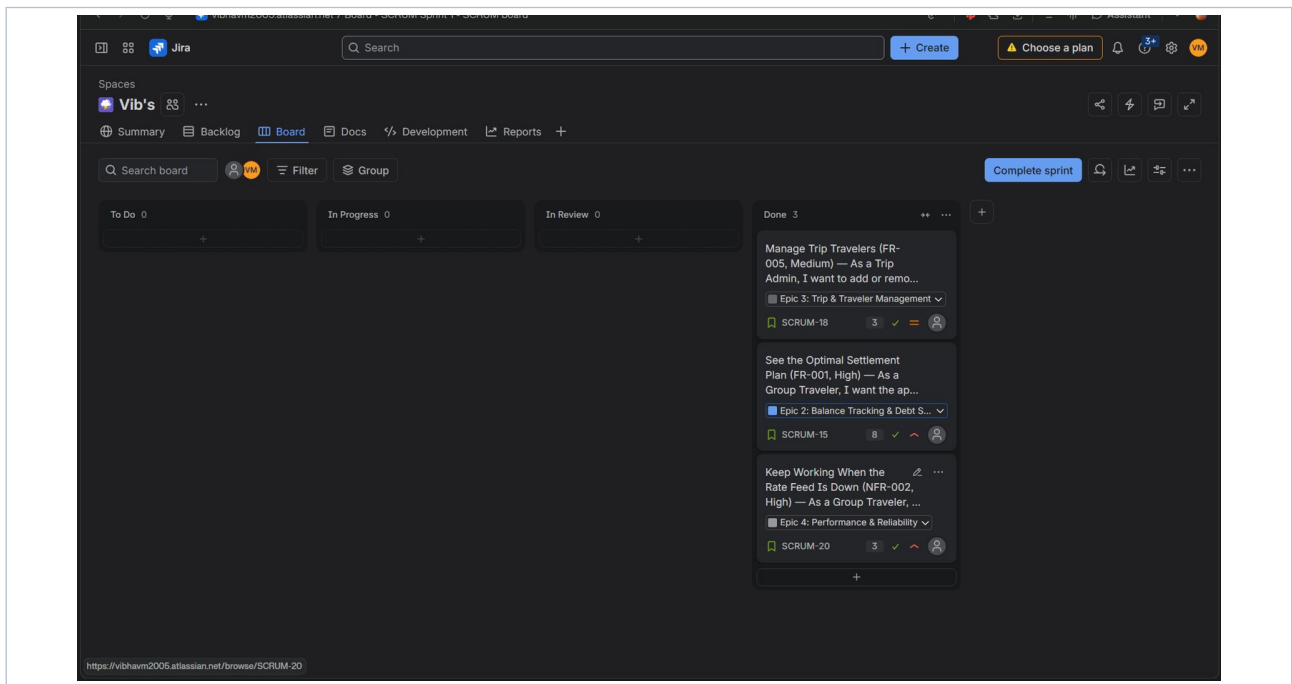
3. Sprint 1 board - work in progress

SCRUM-18 and SCRUM-15 have moved to In Progress while SCRUM-20 is still in To Do, showing that work was picked up incrementally rather than all at once.



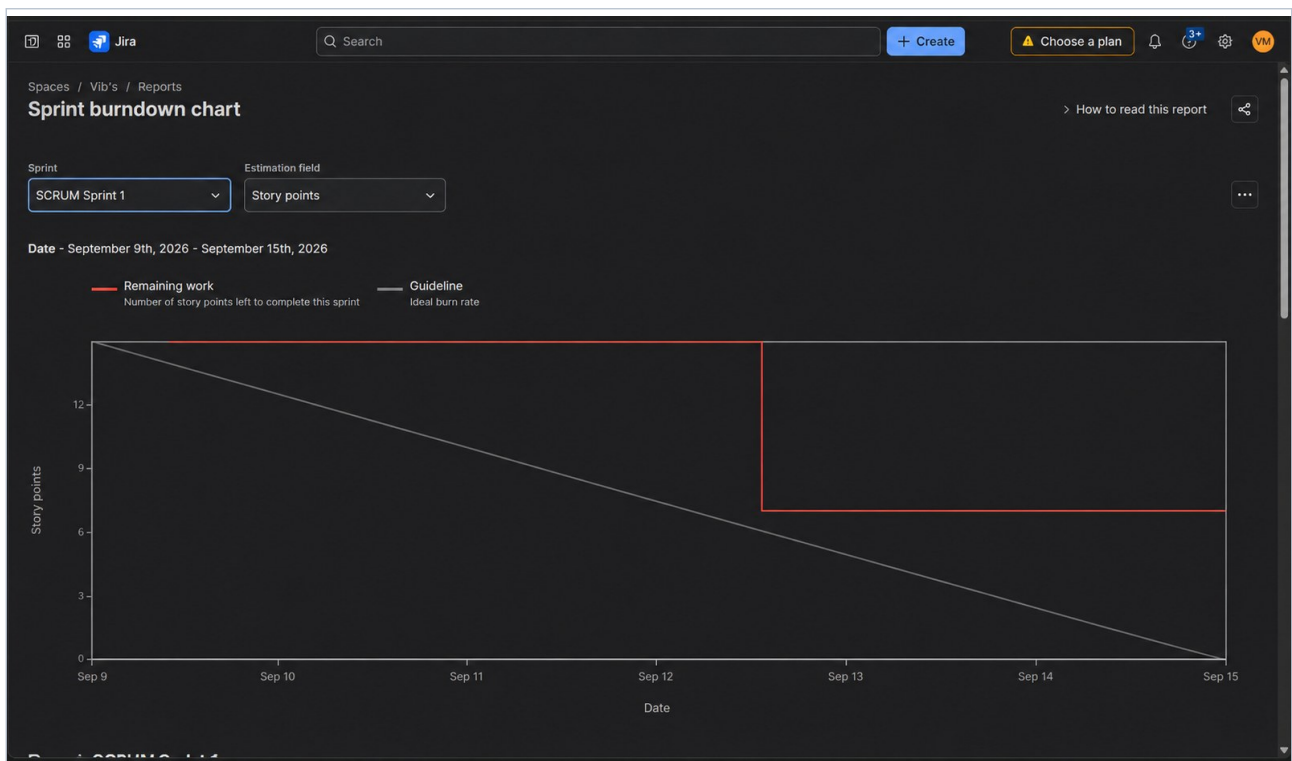
4. Sprint 1 board - completed work

All three sprint stories (SCRUM-18, SCRUM-15 and SCRUM-20) now sit in the Done column and are marked with a completion tick.



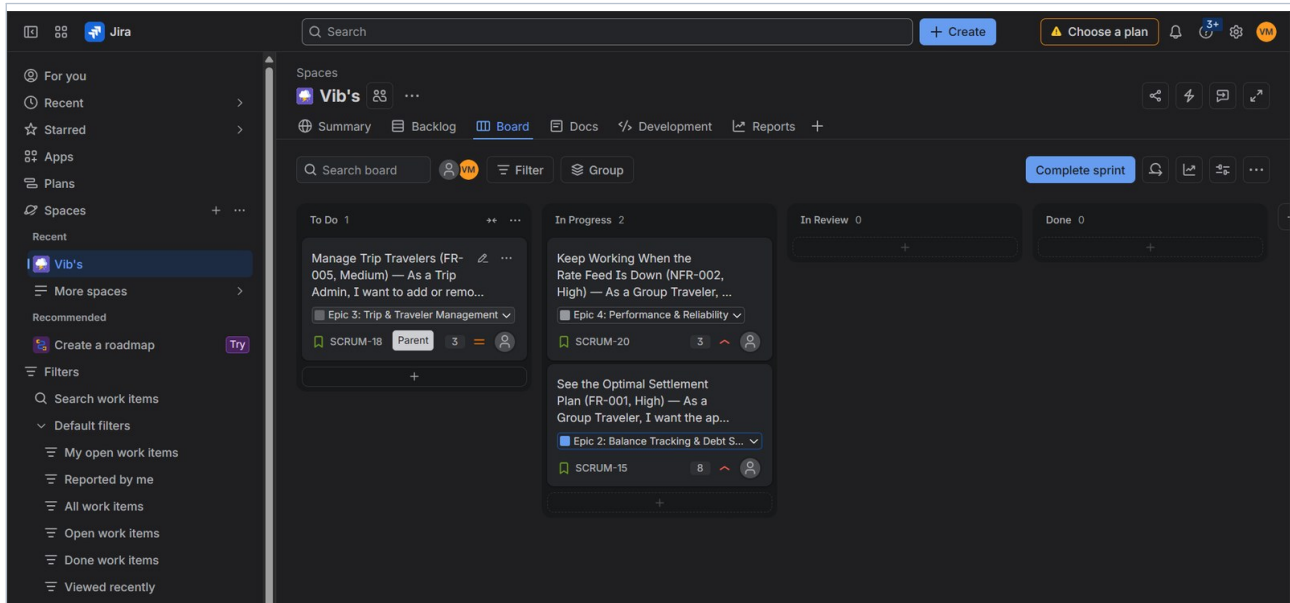
5. Sprint 1 Burndown Chart

Sprint burndown chart for SCRUM Sprint 1 (9 Sep to 15 Sep 2026) using Story points as the estimation field. The red Remaining work line starts at about 15 points and stays flat until around the middle of 12 September, then steps down to about 7 points. The grey Guideline falls steadily from the starting value to 0 on 15 September.



6. Sprint 2 board - work in progress

The Sprint 2 Scrum board: SCRUM-15 (8 pts) and SCRUM-20 (3 pts) are In Progress and SCRUM-18 (3 pts) is still in To Do, with In Review and Done empty. This shows the same To Do to In Progress workflow being applied to the second sprint.



7. Sprint 2 Burndown Chart

Sprint burndown chart for SCRUM Sprint 2, plotted by the hour on 9 September 2026. The Remaining work line starts at about 15 points, drops to about 12 and then to about 7 at roughly 5 PM, and stays there while the Guideline continues to fall towards 0.



6. Sprint and Burndown Analysis

The sprint boards demonstrate the basic Agile feedback loop: a story starts in To Do, moves to In Progress when someone picks it up and reaches Done when it satisfies its acceptance criteria. Recording each state

change in Jira is what feeds the Burndown Chart, so the chart is only as accurate as the board updates behind it.

Sprint 1. The remaining-work line is flat at the top of the chart for the first half of the week and then drops in a single step to about 7 points around 12 September. The guideline, which assumes steady daily progress, stays below the red line for the whole sprint, so the sprint ran behind the ideal pace. About 8 story points (roughly the size of SCRUM-15) were burned in one move, which is characteristic of a large story being closed in one go rather than progress being reported daily.

Sprint 2. The chart is plotted hourly because the simulation was run within a single day. The red line again stays flat for the first part of the sprint, drops to about 12 and then to about 7 points in the late afternoon, and then flattens while the guideline keeps falling. The pattern is the same as Sprint 1: work completes in a few large steps rather than a smooth slope.

The handout notes that a sharp drop in a burndown chart can mean work was not estimated accurately or not broken down finely enough. Here the sharp step is mainly caused by the 8-point story SCRUM-15, which is the clearest candidate for being split into smaller stories (for example graph construction, net-balance calculation and transaction minimisation).

7. Reflection

7.1 Did your estimations reflect the actual effort?

Largely yes, as a relative measure. The Fibonacci scale separated small, well-understood stories (SCRUM-14 and SCRUM-17 at 2 points) from the heavy ones (SCRUM-15 at 8 points for the min-flow settlement algorithm and SCRUM-12 and SCRUM-19 at 5 points). On the board, SCRUM-15 was the story that dominated the burndown, which matches its being the largest estimate. Since this was a simulation, there is no real implementation-time data to prove that the points match actual effort; the one area I would revisit is the 8-point story, which carries the most uncertainty and should be split.

7.2 Was your backlog well-prioritized?

Mostly yes. Priorities followed the Lab 1 rationale: the settlement engine, expense logging, currency conversion, balance view and the two non-functional requirements were High because they define the product, and trip administration was Medium because it supports rather than drives the core flow. One improvement is dependency ordering: the settlement engine (SCRUM-15) needs logged and converted expenses (SCRUM-12 and SCRUM-13) as its input, so in a real project those stories would be scheduled in the same or an earlier sprint than SCRUM-15.

7.3 How did your simulated sprint align with your plan?

The plan was a one-week sprint of 14 points containing three stories, and the board followed that plan: all three stories were taken from To Do through In Progress, and the Done column in Sprint 1 shows the planned work closed on the board. The second sprint reused the same workflow with its own board and burndown chart. The main gap between plan and execution is shape rather than scope: the charts show work finishing in late batches instead of steadily across the sprint.

7.4 What insights did the burndown chart give about your team's capacity?

In both sprints the captured charts show roughly 8 of about 15 points burned, against a plan of 14 to 15, which suggests a realistic capacity nearer 8 points per sprint than 14 for this one-person team. The handout points out that consistently missing the forecast can mean too much was committed; the right response would be to commit fewer points (or split SCRUM-15) and to update the board daily so that the red line follows the guideline more closely. Because the sprints were compressed into a short simulation, a few real sprints would be needed before treating 8 points as a dependable velocity.

8. Conclusion

Lab 2 applied Jira to the Group Travel Expense Simplification Engine started in Lab 1. The seven Lab 1 requirements were turned into four Epics and eight prioritised, estimated User Stories (31 story points), a one-week sprint of 14 points was planned, run across the Scrum board and reviewed through Burndown Charts for two sprints. The exercise showed how requirements traceability, relative estimation and burndown tracking work together, and highlighted two concrete improvements for future sprints: splitting the large settlement story and updating the board continuously so that progress is visible every day.

9. References

1. PES University, Software Engineering Lab 2 handout: *Agile Backlog Creation & Sprint Simulation in Jira*.
2. Lab 1 repository (requirements table, UML use-case diagram and Log Expense use-case flow): github.com/Vibhav1412/SE-LAB1--Group-Travel-Expense-Simplification-Engine
3. Jira workspace: vibhavm2005.atlassian.net, space "Vib's" (key SCRUM).